Introduction to NLP

Param's class, notes might be non existent

NLP → NLU Lang → Comp
⇓ ↳ NLG Comp → Lang
R/W text, Translate, Answer and Summarize

But syntax is not sufficient in probabilistic LLMs
(colourless green ideas ...)

Sentence level context might not be sufficient.
There is also a lot of variation across speakers and communities

What works well (in High Resource Languages)

Pre trained transformers (BERT, GPT)
Clean text → high accuracy on POS, NER, Sentiment Analysis
Translation for pop. lang. pairs.

Manish's class (yay?)

NLP began with Turing and ENIGMA, and got a lot of funding in the Cold War (MT of Soviet data)

Pressures of society fix the medium of comms

language is used for both info transfer and social positioning

Cultures affect language → ambiguity (same word, same sounds)

Memorability?

Principle of Least Effort → Both speaker and listener want to minimise effort

Assumed context → Cooperative Principle
↳ Missing context → Acc.

Diff usecases → Diff levels of completeness
↳ Develop "sublanguages"

Language evolved for generalistic purposes, but can be co-opted for specific purposes.

Challenges in Ambiguity, PP attachment problem, scale

Tokenization and Language Modelling

Scripts are influenced by various thing (context, abjads, etc)

Case markers assign theta roles
Case marking frees word ordering

Ellipsis :- Missing stuff to be understood by context

Elision :- Dropping parts of sentences

One word commands are the shortest sentences
imperative verb

Tokenization :- The task of seperating tokens in a given input sentence
marker in earlier days to count things
one space separated component of a sentence
single surface form word

Type :- Vocabulary word. Might be present in its root form in the dictionary.

Vocabulary :- Set of all Types

Punctuation captures prosody in text

Tokenization :- Identifying tokens in a sentence

Tokens :- Atomic units of text processed by machines

Words are separated from the punctuations but stuff like i'm is preserved

Stemming :- Clipping of suffixes (No base in language studies)
faces → face

↳ Lemma

Type :- Root form ⇒ face and faces → face
Token :- Modified forms ⇒ face & faces

Lemmatization :- Converting a word into it's lemma (base / root form)

All (active) languages have similar diversity in their vocabulary

High TTR (Token : Type Ratio) :- Morphological richness

Challenges :-
(i) Hyphenation
(ii) Numbers
(iii) Dates
(iv) Abbreviations
(v) Punctuations
(vi) URLs
(vii) Emojis (seq of Unicode, naive tokenizers could split these)
(viii) Zero-width joiners

(ix) Code Switching
(x) Disfluency ~~and fragmentation~~
(xi) (a) Fragmentation
(b) Fillers
(xi) Extra punctuation marks for others

~~Mor~~
Morphemes
Minimal unit with meaning in a language
Affix :- Additional context & meaning (prefix / suffix / infix)

Distributional Semantics
Word is meaningless sans context
A word is known by the company it keeps

Inflectional Morphemes :- Adds grammatical sense without changing its class

Derivational Morphemes :- Adds more than grammatical sense changes the class

Clitics :-
Syntactically Independent
Phonologically Dependent (?)

Morphological Typology

Study of how langs vary in morphology, how words break up into their parts

Greenberg's 1960 scale

Analytic ← Synthetic → Polysynthetic

Segmentability

Agglutination & Fusional

Morphemes are complex and require more effort to split them

Data for Tokenization
(i) Morfessor
(ii) Byte Pair Encoding
(iii) Unigram Language Modelling Tokenization

Morphological Typology → How words break into parts

No of morphemes per word
How easily are the morphemes seperatable

Character based tokenization is not sustainable in non-logographic scripts

Embeddings:- Placing objects within the n-dimensional space (of features)
↓
semantic representation space

Convert objects into vectors (n-dimensional)

Similar objects have similar representations

Context helps disambiguate homonymy

Global embedding = 1 embedding / form
Contextual embedding = at runtime

Tokenization free models exist

We can embed whole words, but vocab size sky rockets

Morfessor :-

Minimum Description Length :- What is the smallest explanation of the word.

↳ Keep list of stored morphemes as short as possible

minimizing loss

$$L(S, D) = -(\log(P(S)) + \log(P(D|S)))$$

↳ segmentation → list of roots and words and affixes

$P(S)$ - Prior probability of the lexicon

$P(D|S)$ - Probability of corpus being generated by the morphemes in the lexicon

Tests all possible Two part splits

Generation Modelling and Discriminative Modelling

↓ Modelling $P(D|Y, \theta) \times P(Y)$

↓ Modelling $P(Y|D, \theta)$

Byte Pair Encoding :-

Goes from char → word, repeatedly combines frequent character pairs into symbols

for j in range (# merge operation):
  Get stats of each character bigram in vocab
  Get most frequent bigram (In case of ties → FCFS)
  In vocab merge the most frequent bigram into a unigram

You acquire a Ranked Merge List and to segment

a new unseen word, the model applies the rules in the order of being learnt.

Sentence piece:-

Most frequent bigrams get merged, then remove the least frequent ones.

$x^* = \text{argmax } P(x)$

$P(x) = \prod_{i=1}^{M} p(x_i)$

~~Why the cha~~

Language Modelling:-

Predicting next token ← Give probability distribution

Two threads
↓ ↳ n-gram models
LLM

Word Prediction:-
Query Autocomplete
Spell checking
Grammar checking
Speech Recognition
Rephrasing
Vibe coding

We use joint probability.

$\mathbb{P}(A,B,C,D) = \mathbb{P}(D|A,B,C) \times \mathbb{P}(C|A,B) \times \mathbb{P}(B|A) \times \mathbb{P}(A)$

$$P(w_{1:n}) = \prod_{k=2}^{n} P(w_k \mid w_{1:k-1})$$

How to simplify? Markov Assumption!

$$P(w_n \mid w_{1:n-1}) \approx P(w_n \mid w_{n-1})$$

$\Downarrow$

First Order Markov Model / Bigram Model.

N-grams can't handle long distance dependencies and do poorly at modelling new sequences.

Soln -> Large Language Models because of their use of neural embedding spaces.

First Transformers had a context len of 512 tokens. Now 128,000 tokens on avg.

Reliability vs Discrimination?

Large Green _____

~~see~~ swallowed the large green _____